

Nirosh Kumar

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Summary

- Engineer with over 8+ years of experience specializing in the development, qualification, and production of FDA-regulated medical devices and life sciences.
- Skilled in overall End to End product quality , product testing , SPC , test protocols and plans,quality control plans(incoming inspection),design quality activities,Risk management,APQP,PPAP,supplier qualification
- Skilled in Manufatcuring operations ISO 9001, line setup , minitab, CQV,Process validation(IQ,OQ,PQ),In process Inspection,Equipment validation(furnances,packaging,laser weld machines etc),cGMP,validation protocols,reports ,continous improvement,lean startegy(japanese methods)
- Skilled in Clinical Quality, cGMP methods,CQV,Veevavault
- Skilled in industry standards such as ISO 13485, ISO 14971, IEC 62366,ISO 9000,EUMDR,CGMP,21 CFR Part 820 and FDA regulations.
- Skilled in utilizing tools like Six Sigma,Lean methodology,SolidWorks ,minitab,DOE

Experience

Abiomed | MA, Danvers

Quality Engineer | 01/2024 - Present

- Led process validations (IQ, OQ, PQ) for critical catheter manufacturing processes, including extrusion, laser bonding, tip forming, and coating applications.
- Performed root cause analysis (RCA) and Failure Mode and Effects Analysis (FMEA) to address non-conformances and improve process robustness.
- Designed and implemented in-line process controls (SPC) to monitor key parameters, reducing defect.
- Collaborated with Manufacturing, R&D, and Regulatory teams to implement process improvements and ensure compliance with Good Manufacturing Practices (GMPs).
- Evaluated and qualified raw material suppliers for catheter components, ensuring compliance with ISO 13485 and FDA 21 CFR Part 820.
- Conducted supplier audits and implemented Supplier Corrective Action Requests (SCARs) to address material non-conformities.
- Supported New Product Introduction (NPI) by ensuring design for manufacturability (DFM) and smooth transition from R&D to production.
- Led risk management activities by conducting Design FMEA (DFMEA) and Process FMEA (PFMEA)
- Reviewed and approved engineering change orders (ECOs) to ensure design and process changes met quality and regulatory requirements.
- Ensured compliance with ISO 13485, FDA QSR, and MDR regulations by maintaining rigorous documentation and supporting internal/external audits.
- Led CAPA investigations and NC.
- Supported FDA and Notified Body audits, ensuring audit readiness for 510(k), PMA, and CE Mark approvals.
- Investigated customer complaints and field failures, conducting root cause analysis and implementing corrective actions to enhance product reliability.
- Worked cross-functionally to reduce return rates and improve post-market surveillance for catheter products.
- Developed training programs for production operators to ensure compliance with quality procedures and minimize human errors.

Products Worked : **Impella catheter CP Optical,RP SA ,RP Flex,RP C3, 5.5**

DePuy Synthes (JnJ) | MA, Raynham

Quality Engineer | 06/2020 - 12/2023

- Developed and implemented comprehensive quality control processes, ensuring strict compliance with ISO 13485, FDA and EU MDR standards.
- Led the development and execution of validation and verification protocols, validating product designs, processes, and equipment.
- Implemented proper design controls/documentation to meet FDA 21CFR part 820,EU MDR,DHR,DHF,DMF.

- Risk Assessment : FMEA,CAPA,Risk Potential and Severity,Occurance Rates, Mitigations,V&V activities.
- Cross Functional support : RA,R&D, Product Development,Supplier Quality, NPI,PMS,Marketing.
- Standards known : ISO14001, ISO13485,ISO14971, ISO 9001, IPC-A-610, IEC62304,IEC 62366,ISO 11607-1
- Implement CAPA,NCMR,Post Market Surveillance activities
- Customer /Supplier Communication , Audit participation
- Supporting design control activities for NPD efforts and performed gap assessment (RMP, RAC, DFMEA, RMR, IOVV) for EUMDR 2017/745)

Products Worked : **Expedium Spine, Expedium Fenestrated screws and plates , Expedium Instruments, Viper prime ,Symphony, Trialtis , concord**

GE Healthcare Life Sciences (Cytiva) | MA, Westborough Manufacturing Engineer | 02/2018 - 06/2020

- Utilized regulations 21CFR 820.100 (CAPA), 21CFR 820.198 (Complaint files), 21CFR 820.90 (NC's), ISO 13485:2016,ISO 14971 in different manufacturing areas like ISO-13485, 21.CFR.PART11 and implementation of changes to Standard Operating Procedures, Work Instructions.
- Utilized drawing prints to reduce redundancy, performing time studies to reduce total operation time (analyzing workflow) and improving procedures accordingly.
- Successfully performed Gage Repeatability and Reproducibility (Gage R&R using MINITAB) to improve the measurement system for multiple products.
- Performed Commisioning , Qualification and Process Validations (IQ/OQ/PQ)
- Implemented accelerated aging , tensile test , break test, leak test , seam strength test on bioreactor bags for used for cell expansion .
- Troubleshoot process maps, control plans and value stream mapping.Drove continuous improvements in process design and operational performance.
- Evaluated new equipment/processes for productions operations. worked with quality to conduct supplier audits and onboard suppliers.
- Application od Six Sigma tools such as Cause and Effect Diagrams (Fishbone diagrams), Pareto analysis, and Failure Mode and Effects Analysis (FMEA) to drive process improvements.
- Expert in: Kanban, 5S, Six sigma , Lean Methodology and SPC.
- Training technicians and operators on process procedures.

Products Worked: **XDR 10 , ready to process cellular Mixer Bags, XURI basic cell bags , XURI classic perfusion filter bags**

Flex Medical | TX, Irving Quality Assurance Engineer | 02/2017 - 02/2018

- Implemented business processes such as Engineering Change Order (ECO) Process, NCMR Process and Design for Manufacturability (DFM), Design for Assembly (DFA) input to improve and enhance manufacturability.
- Implemented product development of NPI X ray machine for client . Implemented characterization, feasibility, design & development , design review , design transfer phases.
- Oversaw quality for an X-ray machine prototype for osteoporosis from user requirements to testing.
- Oversaw material disposition, deviations , NCMR and MRB.
- Supplier audits , Supplier qualification and supplier on board process
- Train technicians and operators in material receiving and FAI .
- Led CAPAs and resolved NCMRs . Part of QRB on key decisions .
- Led Test Method Validations, developed protocols and reports.
- Generated QMS ISO 13485 procedures and ensured compliance with procedures and documents, guiding management.
- worked with software team in SDLC , resolving issue related to design control and develop software quality documents.
- Drafted detailed FMEA reports, communicating risks and mitigation strategies.
- Collaborated with electrical(PCBA),design,Program Management,Regulatory, Lab , supplier , NPI and validation team to ensure documentation.
- Ensured correct performance of PLC gate, design, and technical reviews for each project phase and ensured build readiness for various project stages

Product Worked: **Horizon DXA system from Hologics**

Baxter | CO, Englewood Quality Engineer | 07/2016 - 02/2017

- Managed investigations of non-conforming products/processes, compiling technical reports for NCMR investigations,
- Maintained Risk Management File in the electronic Design History File.

- Implemented and managed Design History File, design requirements, V&V, and risk management analysis.
- Facilitated risk analysis meetings to evaluate failure modes and implement risk control measures.
- Developed and executed Test Method Validation protocols and Gage R&R studies

Products Worked : **Actifuse , Altapore , Surgical Tools**

GE Healthcare | KA, Bangalore

Manufacturing Quality Engineer | 05/2013 - 11/2014

- Improved production efficiency by implementing Lean Manufacturing practices and reducing cycle times.
- Validated manufacturing processes and equipment through IQ, OQ, and PQ to meet quality standards.
- Optimized assembly processes for healthcare devices, ensuring safety, quality, and cost-efficiency.
- Collaborated with cross-functional teams to ensure smooth product transitions from design to production.
- Conducted time studies to identify bottlenecks and enhance production workflows.
- Designed and validated tooling and fixtures to improve manufacturability and reduce operator errors.
- Performed root cause analysis to resolve defects and improve first-pass yield.
- Created detailed work instructions and process documentation for regulatory compliance.
- Led Six Sigma projects to minimize process variations and enhance product quality.
- Trained operators and technicians on assembly processes, equipment use, and quality standards.
- Monitored manufacturing metrics to drive continuous process improvements.
- Supported new product introductions by ensuring manufacturability and scalable processes.
- Reduced costs through waste elimination and process automation initiatives.
- Contributed to risk management by performing FMEA and improving control plans.
- Resolved production issues to minimize downtime and meet delivery schedules.
- Designed layouts to optimize floor space and improve manufacturing efficiency.

Skills

MDD/MDR (Expert), ISO13485 (Expert), Drug delivery (Expert), Drug elution (Expert), Testing (Experienced), Risk management (Expert), Design history file management (Expert), AutoCAD (Experienced), SolidWorks (Experienced), SolidWorks Simulation (Skillful), Quality improvement strategy (Expert), Root cause analysis (Expert), Change management (Experienced), Risk mitigation (Expert), Non-conformance processes (Expert), Corrective action planning (Expert), Design and manufacturing reviews (Expert), QMS 13485 support (Expert), FDA regulations compliance (Expert), Validation protocols, Medical device regulations

Education

New England Collage | Henniker
MBA | 05/2021

Pittsburg State University
Engineering Technology | 05/2016

Certificates

Six Sigma Certification, CQE(Pursuing)